

RG04

## Natural Vegetation and Forests of Rajasthan

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*Two to three questions a paper within the 11-15 Rajasthan Geography block. State tree and state flower, Dhok as the commonest species, Amrita Devi and the 363, forest cover percentage, and sacred groves are the recurring items. Match-the-following is the dominant format.*

Rajasthan is India's largest state and one of its least forested. That contradiction is the whole chapter. RPSC asks it in three ways - the ISFR numbers, the species-to-botanical-name match, and the Khejarli story. Learn the numbers with their report edition attached, because a figure without an edition is a wrong answer waiting to happen.

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## Forests of Rajasthan

### The numbers (ISFR 2023)

- Forest cover 16,548.21 sq km
- 4.84 per cent of state area
- Forest + tree cover 8.00 per cent
- Open forest dominates
- India's figure 21.76 per cent

### Legal classes

- Recorded forest area about 32,700 sq km
- Protected Forest about 56 per cent
- Reserved Forest about 38 per cent
- Unclassed about 6 per cent
- Rajasthan Forest Act, 1953

### Forest types

- Group 5 Tropical Dry Deciduous about 57 pc
- Group 6 Tropical Thorn about 15 pc
- Northern Dry Mixed Deciduous - largest single type
- Sub-tropical evergreen only at Mount Abu
- Halophytes at Sambhar, Pachpadra, Didwana

### Key species

- Dhok - most widespread
- Palash - second most numerous
- Khejri - state tree
- Rohida - state flower
- Teak, Salar, Khair, Bamboo
- Kair, Aak, Phog, Sewan

### Districts

- Most forest by area - Udaipur
- Highest share - Udaipur, Pratapgarh
- Then Sirohi, Karauli, Baran
- Least - Jodhpur, Churu, Jaisalmer, Bikaner

### Conservation tradition

- Khejarli 1730 - Amrita Devi, 363 Bishnois
- Bishnoi sect, Jambhoji 1485
- Orans and other sacred groves
- 11 September - Forest Martyrs Day

### Policy and projects

- Forest Policy 2010, Forest Policy 2023
- 20 per cent green cover target
- Joint Forest Management - VFPMCs
- Aravalli Green Wall Project, March 2023
- CAZRI and AFRI, Jodhpur

**The numbers - and always name the report edition**

The Forest Survey of India publishes the India State of Forest Report every two years. Rajasthan's figures move each edition, so a number is only useful when you can say which report it came from. The set below is ISFR 2023, released in December 2024 - the edition your standard sources are built on. Quote these, and quote the edition with them.

- Forest cover of Rajasthan - 16,548.21 sq km, which is 4.84 per cent of its geographical area (ISFR 2023).
- Forest plus tree cover - 27,389.33 sq km, that is 8.00 per cent of the state (ISFR 2023).
- India's national forest cover in the same report is 21.76 per cent - Rajasthan is far below it.
- Rajasthan gained about 394 sq km of forest and tree cover over 2021 - the fourth-largest increase among Indian states; Chhattisgarh led with 684 sq km.

### Canopy density classes in Rajasthan (ISFR 2023)

| Class                   | Area (sq km) | Point to note                                                  |
|-------------------------|--------------|----------------------------------------------------------------|
| Very Dense Forest       | 223.20       | Tiny - Rajasthan has almost no closed canopy                   |
| Moderately Dense Forest | 4,237.41     | The middle tier                                                |
| Open Forest             | 12,087.60    | Overwhelmingly dominant - about three-quarters of forest cover |
| Total forest cover      | 16,548.21    | 4.84 per cent of the state                                     |

#### TRAP

Forest COVER and RECORDED FOREST AREA are two different numbers. Forest cover is what satellites see - 16,548 sq km, 4.84 per cent. Recorded forest area is what is legally notified as forest - about 32,700 sq km, roughly 9.6 per cent. Almost double. RPSC swaps them.

**CURRENT LINK**

Current link (as of September 2026): the ISFR is biennial, so a newer edition may already be on the table by exam day. Treat 4.84 per cent as the ISFR 2023 figure and check the Rajasthan row of the latest FSI report before December 2026. In an exam, if the question names a report year, answer for that year's figure.

## Recorded forest area and the legal classes

Legally, forests are of three classes. Reserved Forest is the strictest - everything is prohibited unless permitted. Protected Forest is the looser class - everything is permitted unless prohibited. Unclassed is the residue. Rajasthan is unusual in that Protected Forest, not Reserved, is its largest class.

**Legal classes of Rajasthan's recorded forest area (about 32,700 sq km, roughly 9.6 per cent of the state)**

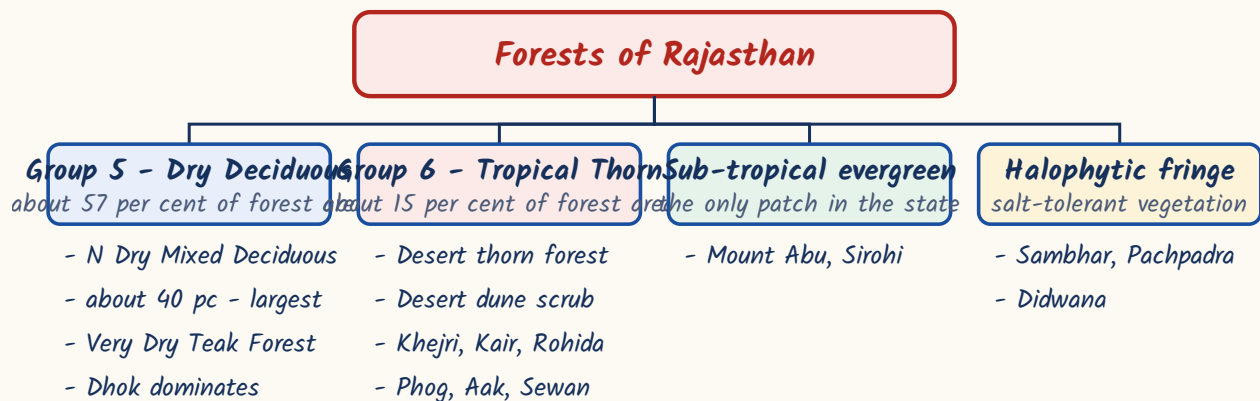
| Class            | Share             | Character                                                          |
|------------------|-------------------|--------------------------------------------------------------------|
| Protected Forest | About 56 per cent | Largest class in Rajasthan; rights allowed unless expressly barred |
| Reserved Forest  | About 38 per cent | Strictest protection                                               |
| Unclassed Forest | About 6 per cent  | Residual category                                                  |

- The governing state statute is the Rajasthan Forest Act, 1953.
- Joint Forest Management is an administrative and participatory arrangement layered on top of that Act - it does not replace it.

## Forest types - Champion and Seth applied to Rajasthan

Champion and Seth classified Indian forests into sixteen groups. Rajasthan's forests fall almost entirely into two of them - Group 5, Tropical Dry Deciduous, and Group 6, Tropical Thorn. Group 5 dominates the Aravallis and the east; Group 6 takes over as you move west into the desert. Two more slivers exist: a sub-tropical evergreen patch on Mount Abu, and salt-tolerant vegetation around the saline basins.

### Champion and Seth applied to Rajasthan



### Forest types matched to their districts

| Type                             | Share / status                         | Where in Rajasthan                                                                   |
|----------------------------------|----------------------------------------|--------------------------------------------------------------------------------------|
| Group 5 - Tropical Dry Deciduous | About 57 per cent of forest area       | Aravalli belt and eastern hills - Udaipur, Alwar, Chittorgarh, Bundi, Sawai Madhopur |
| Northern Dry Mixed Deciduous     | Largest single type, about 40 per cent | Dhok-dominated Aravalli and eastern hill tracts                                      |
| Very Dry Teak Forest             | Sub-type of Group 5                    | Banswara, Udaipur, Chittorgarh, Pratapgarh, Dungarpur, Baran                         |
| Group 6 - Tropical Thorn         | About 15 per cent of forest area       | Arid west - Jaisalmer, Barmer, Bikaner, Churu, Jodhpur, Nagaur                       |

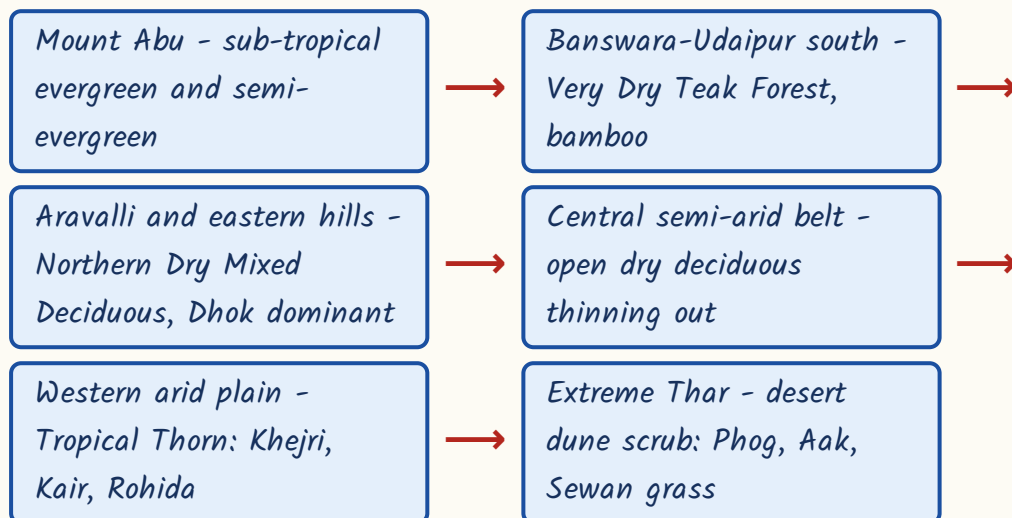
| Type                                    | Share / status          | Where in Rajasthan                         |
|-----------------------------------------|-------------------------|--------------------------------------------|
| Sub-tropical evergreen / semi-evergreen | Only patch in the state | Mount Abu tract, Sirohi district           |
| Salt-tolerant (halophytic) vegetation   | Fringing type           | Sambhar, Pachpadra and Didwana salt basins |

- Thorn forest is dominated by Khejri, Kair, Rohida, Phog and Aak.
- Desert thorn forest and desert dune scrub are sub-types within Group 6.
- Teak and bamboo belong to the moister dry deciduous forests of the south - never to the thorn belt.

### TRAP

Rajasthan has NO tropical wet evergreen forest. Its only evergreen element is the sub-tropical evergreen patch at Mount Abu. Any option offering wet evergreen forest for Rajasthan is the incorrect statement.

**How the vegetation changes as you walk from the wettest corner to the driest**



**The species - this is where the matching marks are**

Learn each tree with three tags: local name, botanical name, and the one thing it is famous for. RPSC's matching questions pair any two of the three. Dhok is the one to fix first - it is both the most widespread species and the most numerous by tree count.

### Where each species and type belongs



the rain - evergreen only on Mount Abu, teak and bamboo in the south, dhok on the Aravallis, thorn and grass in the west

### Species of Rajasthan - local name, botanical name, what it is known for

| Local name    | Botanical name            | Known for                                                                           |
|---------------|---------------------------|-------------------------------------------------------------------------------------|
| Dhok          | <i>Anogeissus pendula</i> | Most widespread and most numerous species; dominates the Aravalli and eastern hills |
| Palash / Dhak | <i>Butea monosperma</i>   | 'Flame of the Forest'; second most numerous species; lac culture, leaf plates, gum  |
| Khejri        | <i>Prosopis cineraria</i> | State tree; 'Kalp Vriksha of the desert'; also called Shami or Jandi                |
| Rohida        | <i>Tecomella undulata</i> | State flower; 'Marwar Teak' or 'Desert Teak'; prized timber of the arid west        |

| Local name       | Botanical name                 | Known for                                                               |
|------------------|--------------------------------|-------------------------------------------------------------------------|
| Teak<br>(Sagwan) | <i>Tectona grandis</i>         | Very Dry Teak Forest of the humid south                                 |
| Salar            | <i>Boswellia serrata</i>       | Dry rocky Aravalli slopes; yields Indian frankincense resin             |
| Khair            | <i>Acacia catechu</i>          | Riverine and ravine tracts; yields katha (catechu) for paan and tanning |
| Bamboo<br>(Bans) | <i>Dendrocalamus strictus</i>  | Udaipur, Banswara, Dungarpur, Chittorgarh, Pratapgarh, Kota, Baran      |
| Kair             | <i>Capparis decidua</i>        | Desert shrub; its berries make 'ker' in the ker-sangri dish             |
| Aak              | <i>Calotropis procera</i>      | Common desert milkweed of waste and dune land                           |
| Phog             | <i>Calligonum polygonoides</i> | Dune-binding shrub of the Thar                                          |
| Sewan grass      | <i>Lasiurus indicus</i>        | Finest natural pasture grass of the Jaisalmer plains                    |
| Ber              | <i>Ziziphus mauritiana</i>     | Cultivated fruit; the wild bush Jharber is <i>Ziziphus nummularia</i>   |

**ASKED IN RAS**

Asked and re-asked - which is the most widespread tree species in the forests of Rajasthan? Answer: Dhok, *Anogeissus pendula*. The botanical name is asked as often as the common name.

**YAAD RAKHO**

Remember the product pairs: Khair gives KATHA (both start with K-A), Salar gives resin (frankincense), Kair gives KER berries, Khejri gives SANGRI pods. Ker plus sangri is the dish - so Kair plus Khejri is the pair.

- Palash is valued for flowers, gum, leaf plates and lac culture - NOT for commercial sleeper timber; that is the standard wrong pairing.
- Khejri is a leguminous tree of the arid plains, never a conifer - another planted error.
- Teak and bamboo are southern species; pairing either with Jaisalmer or Barmer is always wrong.

## State tree and state flower - a two-line certainty

Two questions in this chapter are as near to certain as anything in the paper. Khejri is the state tree. Rohida is the state flower. Both are trees of the arid west, which is exactly why RPSC pairs them with humid-south districts in the wrong options.

### Khejri versus Rohida

| Khejri ( <i>Prosopis cineraria</i> )                                | Rohida ( <i>Tecomella undulata</i> )                               |
|---------------------------------------------------------------------|--------------------------------------------------------------------|
| State TREE of Rajasthan                                             | State FLOWER of Rajasthan                                          |
| Called Kalp Vriksha of the desert                                   | Called Marwar Teak or Desert Teak                                  |
| Also known as Shami or Jandi                                        | Valued chiefly for its durable timber                              |
| Pods = sangri (eaten as vegetable); dried leaves = loong (fodder)   | Bright orange-red flowers of the arid west                         |
| Backbone of desert agro-forestry - grows in the field with the crop | A tree of the arid and semi-arid west, not of the humid south-east |

#### ASKED IN RAS

Both halves asked separately in past papers - which tree is the State Tree of Rajasthan? Khejri. Which is the State Flower? Rohida. Learn the botanical names too; they appear as options.

- Khejri is a leguminous tree of the arid plains - never a conifer, whatever an option suggests.
- It is the mainstay of the desert agro-forestry system: fodder, fuel, shade, edible pods, nitrogen for the soil.
- Ker (Kair berries) plus sangri (Khejri pods) makes ker-sangri, Rajasthan's best-known dry-land dish.

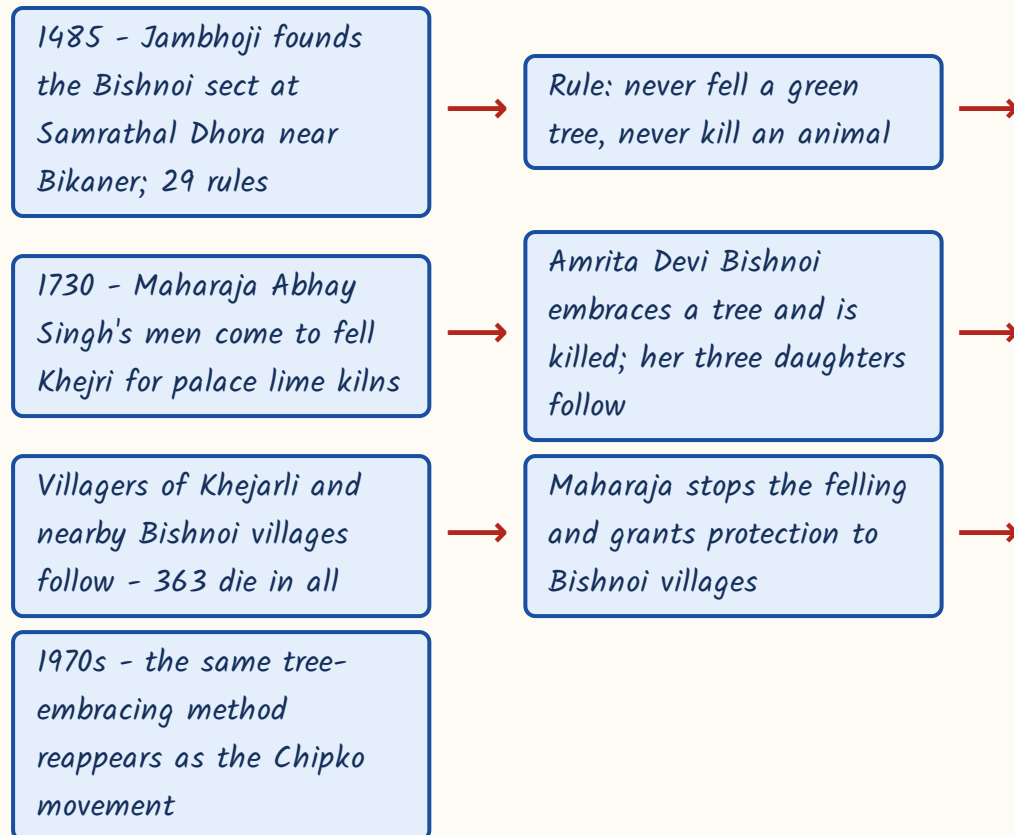
### **Khejarli, 1730 - learn this as a story, not a bullet**

In 1730 the Maharaja of Marwar, Abhay Singh, needed lime for a new palace at Jodhpur. Lime is burnt in kilns, kilns need firewood, and the nearest firewood was the Khejri groves around the village of Khejarli. The king's men went out with axes. The village was Bishnoi - a sect whose religious code forbids the felling of a green tree.

A woman named Amrita Devi Bishnoi put her arms around a Khejri and told the axemen they would have to cut through her first. They did. Her three daughters followed her. Then the villagers of Khejarli and the surrounding Bishnoi villages came, one after another, and embraced the trees. When the killing stopped, 363 Bishnois were dead. The Maharaja, on hearing what had happened, halted the felling and issued an order protecting Bishnoi villages from tree-cutting and hunting.

Two reasons to know this properly rather than as a line. First, RPSC asks it almost every cycle - the leader, the number, the ruler, the year. Second, it is the acknowledged ancestor of the Chipko movement of the 1970s: the same method, the same idea, two and a half centuries earlier.

### Khejarli - the sequence



#### ASKED IN RAS

Asked in multiple forms - who led the Khejarli sacrifice of 1730? Amrita Devi Bishnoi. How many Bishnois died? 363. Under which ruler? Maharaja Abhay Singh of Marwar (Jodhpur).

- 11 September is observed as National Forest Martyrs Day in memory of Khejarli and of all who died protecting forests.
- The Amrita Devi Bishnoi Wildlife Protection Award honours exemplary courage in protecting wildlife and forests.
- Bishnoi = 'bis' plus 'noi' = twenty-nine, the number of rules laid down by Guru Jambheshwar (Jambhoji) in 1485.
- Khejarli village lies in Jodhpur district.

**YAAD RAKHO**

Remember: 29 rules, 1485 founding, 1730 sacrifice, 363 martyrs, 11 September remembrance. Five numbers, one story.

## Orans and the sacred grove tradition

An oran is a patch of forest set aside in the name of a local deity and protected by the village itself. No felling, often no grazing, sometimes not even the removal of a dead branch. Orans have preserved biodiversity in the driest districts precisely where the Forest Department has almost no notified forest. They are the community half of Rajasthan's conservation story.

- The names for sacred groves in Rajasthan: Oran, Kankad bani, Deo bani, Rakhat bani, Malvan.
- Orans are common in the western and Aravalli districts and are usually attached to a temple or a folk deity.
- They often enclose the only surviving old-growth trees and the only reliable village water source.
- Many orans of Jaisalmer are critical habitat for the Great Indian Bustard.

**CURRENT LINK**

Current link (as of September 2026): in December 2024 the Supreme Court directed that orans be treated as deemed forests, which brings forest-conservation protection to oran land. This is a fresh, dateable fact of exactly the kind RPSC lifts for a current-affairs question - remember the month, the year and the phrase 'deemed forest'.

**TRAP**

Khadin is NOT a sacred grove. It is a traditional runoff-farming and water-harvesting structure of Jaisalmer. In a list of grove names - Oran, Kankad bani, Deo bani, Rakhat bani - khadin is the odd one out.

## Districts - which have the most forest and which the least

District-level forest data comes from the ISFR district tables, and the widely used set for Rajasthan is ISFR 2021. Note the edition when you quote a district figure. There are two separate rankings and you must not mix them - largest forest by AREA, and highest forest as a SHARE of district area.

### Most forest and least forest, district by district (ISFR 2021)



Green - the forested south and the Alwar hills; red - the bare arid and canal-irrigated north-west

### Forest cover by district (ISFR 2021 - state the edition when you use these)

| Ranking                                        | Order                                                                                                           | Note                                              |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Largest forest cover by area                   | Udaipur (about 2,753 sq km), then Alwar (about 1,196), Pratapgarh (about 1,034), Baran, Chittorgarh (about 990) | Udaipur leads on both counts                      |
| Highest forest cover as share of district area | Udaipur (about 23.5 per cent), Pratapgarh (about 23.2 per cent), then Sirohi, Karauli, Baran                    | Udaipur and Pratapgarh are nearly level           |
| Lowest forest cover percentage                 | Jodhpur, Churu, Jaisalmer, Bikaner, Hanumangarh, Nagaur                                                         | All in the arid or canal-irrigated west and north |

- Trick for the top five by area: U-A-P-B-C - Udaipur, Alwar, Pratapgarh, Baran, Chittorgarh.
- Alwar is the only northern district in that list; every other one is southern or south-eastern.
- Udaipur leads both rankings, by area and by share - so it is the safe answer when the stem is vague.

Watch for Churu. It is among the lowest forest-cover districts in the state, under one per cent of its area, and it is regularly planted into a highest-forest-cover option set to catch the careless. Jodhpur is the same kind of trap from the other direction - a very large district with very little forest.

- The district map changed in 2023 and again in December 2024; Rajasthan now has 41 districts and 7 divisions.
- ISFR district tables and the standard textbook lists still use the older district names - answer in the names the question uses.

## Policy, JFM and greening the desert

The national forest policy target is 33 per cent of geographical area under forest and tree cover. Rajasthan, at 8 per cent, has openly accepted that 33 is out of reach and set itself a lower, honest target. That target number is asked.

- Rajasthan's forest policy target - about 20 per cent of geographical area under forest and tree cover.
- Rajasthan Forest Policy of 2010, restated in the Rajasthan Forest Policy 2023.
- Joint Forest Management in Rajasthan runs through Village Forest Protection and Management Committees.
- JFM provides for sharing of usufruct - fodder, fuel, minor forest produce - with the participating village.
- Social forestry raises trees outside notified forest - on roadsides, canal banks, village commons and farm bunds - to take pressure off the forests themselves.

Desert afforestation is a distinct technical programme, not ordinary plantation. The Thar's problem is moving sand, so the work is to fix the dune first and plant afterwards. Shelterbelts are raised along the Indira Gandhi Nahar to stop dunes from marching into the canal and silting it up. The species used are the hardy natives - Khejri, Rohida, Phog, Sewan grass.

#### Four ways Rajasthan puts trees back



- Shelterbelt plantation along the Indira Gandhi Nahar - checks dune movement and canal silting.
- Sand-dune stabilisation using micro-windbreaks and hardy species.
- The Desert Development Programme operates in the arid western districts, not in the eastern plain.
- CAZRI (Central Arid Zone Research Institute) and AFRI (Arid Forest Research Institute), both at Jodhpur, are the lead research bodies.

### CURRENT LINK

Current link (as of September 2026): the Aravalli Green Wall Project, launched in March 2023, aims to green a 5-km buffer along the Aravalli range across four states - Rajasthan, Haryana, Gujarat and Delhi. Its object is to check land degradation and the eastward creep of the Thar, and it forms part of India's wider land-restoration commitments. Remember the month and year of launch and the four states; that is the examinable core.

### REVISION RECAP

1. ISFR 2023: Rajasthan forest cover 16,548.21 sq km = 4.84 per cent; forest plus tree cover 8.00 per cent; India 21.76 per cent.
2. Canopy split (ISFR 2023): VDF 223.20, MDF 4,237.41, Open Forest 12,087.60 sq km - open forest dominates.
3. Rajasthan gained about 394 sq km of forest and tree cover over 2021 - fourth-largest gain; Chhattisgarh first with 684.
4. Recorded forest area about 32,700 sq km (9.6 per cent) - Protected 56 pc, Reserved 38 pc, Unclassed 6 pc. Governing law: Rajasthan Forest Act, 1953.
5. Champion and Seth: Group 5 Tropical Dry Deciduous about 57 pc, Group 6 Tropical Thorn about 15 pc.
6. Northern Dry Mixed Deciduous is the single largest forest type, about 40 per cent of forest area.
7. Sub-tropical evergreen forest occurs only at Mount Abu; no tropical wet evergreen forest anywhere in the state.
8. Halophytic vegetation rings Sambhar, Pachpadra and Didwana.

9. Dhok (*Anogeissus pendula*) - most widespread and most numerous; Palash (*Butea monosperma*) - second, 'Flame of the Forest'.
10. Khejri (*Prosopis cineraria*) = state tree, Kalp Vriksha, Shami/Jandi, pods sangri, leaves loong.
11. Rohida (*Tecomella undulata*) = state flower, Marwar Teak, arid west.
12. Khair gives katha, Salar gives frankincense resin, Kair gives ker berries, Sewan (*Lasiurus sindicus*) is the Jaisalmer pasture grass, Phog (*Calligonum polygonoides*) binds dunes.
13. Bamboo is *Dendrocalamus strictus*, in Udaipur, Banswara, Dungarpur, Chittorgarh, Pratapgarh, Kota, Baran.
14. Khejarli 1730: Amrita Devi Bishnoi and 363 Bishnois killed resisting Maharaja Abhay Singh's men; ancestor of Chipko.
15. Bishnoi sect - Jambhoji, 1485, Samrathal Dhora near Bikaner, 29 rules; 11 September = National Forest Martyrs Day; the Amrita Devi Bishnoi award is for wildlife protection.
16. Sacred groves: Oran, Kankad bani, Deo bani, Rakhat bani, Malvan. Khadin is a water structure, not a grove.
17. December 2024 - Supreme Court directed that orans be treated as deemed forests.
18. Most forest by area: Udaipur, Alwar, Pratapgarh, Baran, Chittorgarh (ISFR 2021). Least: Jodhpur, Churu, Jaisalmer, Bikaner.
19. State forest policy target about 20 per cent green cover (national goal is 33 pc); JFM runs through Village Forest Protection and Management Committees.
20. Aravalli Green Wall Project - March 2023, 5-km buffer, four states. CAZRI and AFRI at Jodhpur lead arid-zone research.